Bahria University
Discovering Knowledge

Complex Computing ACTIVITY

Submitted to: MISS AMNA IFTIKHAR

Name	Enrollments
Inaamallah Sarfraz	*02-136232-049*
Syed Haseeb Aamir	*02-136232-012*
Sameer Siddiqui	*02-136232-046*

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Smart Waste Classifier – Object Detection System

A Deep Learning–based Waste Segregation Solution

1. Abstract

This project introduces the **Smart Waste Classifier**, a cutting-edge, deep learning–powered system developed to automate the detection and classification of waste items in real-time. Utilizing the YOLOv5 object detection model and leveraging the power of computer vision, this system can accurately distinguish between various types of waste, including **recyclable, non-recyclable, and compostable materials**. The classifier has been trained to identify common waste items such as plastic bottles, diapers, glass, paper, and more, thus enabling more efficient waste management and environmental sustainability.

2. Introduction

Effective waste management is a cornerstone of **sustainable urban living**. One of the most critical steps in this process is **waste segregation**, which ensures that recyclable and compostable materials are properly separated from general waste. Manual segregation is time-consuming, error-prone, and labor-intensive.

To address these challenges, we present the **Smart Waste Classifier**, an innovative system that automates waste detection and classification in real time using computer vision and deep learning. The system is designed to be **versatile and user-friendly**, supporting both webcam-based and image upload inputs. It can be easily deployed as a web-based application, making it accessible for households, industries, and smart city waste management systems.

3. Objectives

- Accurately classify waste into recyclable, non-recyclable, and compostable categories.
- Utilize state-of-the-art deep learning techniques (YOLOv5) for real-time object detection.
- Provide confidence scores and clear recyclability labels for each identified waste item.
- Support both webcam and static image uploads for versatile deployment.
- Deploy the system with an intuitive Streamlit-based web interface for easy access and interaction.

4. System Requirements

Software Requirements:

- Python 3.10+
- PyTorch (for YOLOv5 model)
- OpenCV (for image processing and camera handling)
- Streamlit (for web UI)
- YOLOv5 repository (pre-trained weights and inference pipeline)

IDE / Development Tools:

- Jupyter Notebook / Google Colab

5. Technologies Used

Frontend: Streamlit
Backend: Python (Torch, OpenCV)
Model: YOLOv5
Deployment: Google Colab / Streamlit

6. System Design

Architecture:

The system follows a **modular architecture** with the following components:

- **Model Loader:**
  Loads the pre-trained YOLOv5 model weights and configures the detection pipeline.
- **Controller:**
  Handles image input (webcam or file), preprocessing, model inference, and post-processing.
- **Streamlit UI:**
  Provides an interactive interface for the user to upload images, trigger detection, and view annotated results.

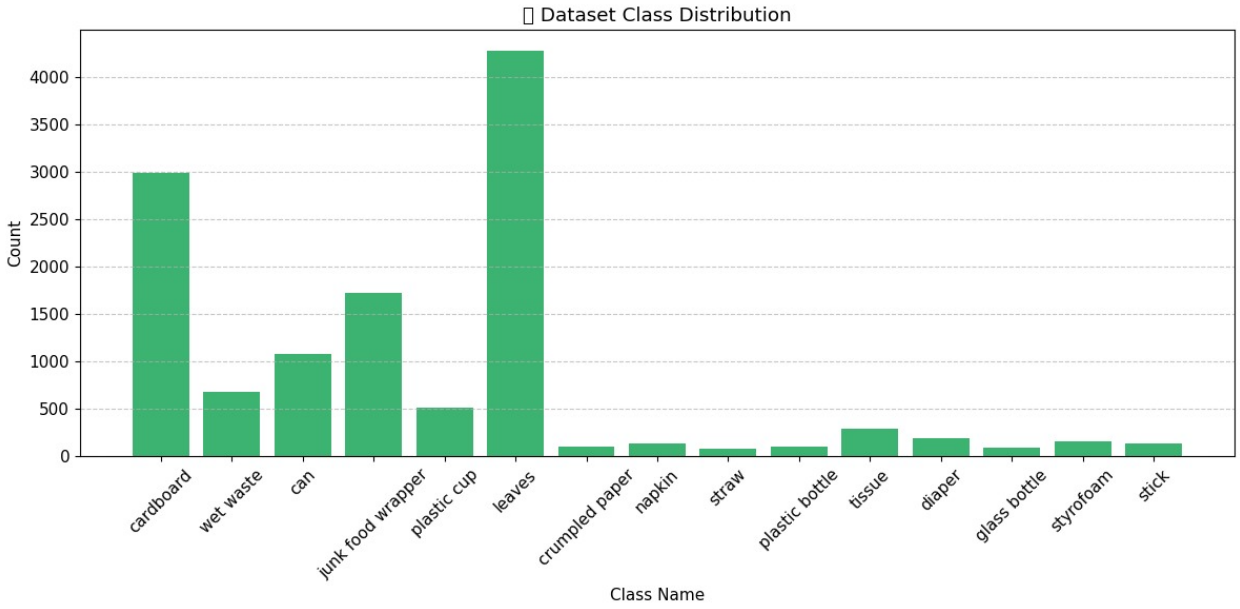
7. Implementation

- The **frontend** is built using **Streamlit**, enabling real-time, interactive usage via a web browser.
- The **backend** leverages **YOLOv5** for object detection and **OpenCV** for image processing (e.g., resizing, color conversion).
- Inference results include **bounding boxes**, **labels**, and **confidence scores**, which are rendered onto the original image.
- The system supports **downloading of annotated images** for later analysis or reporting.

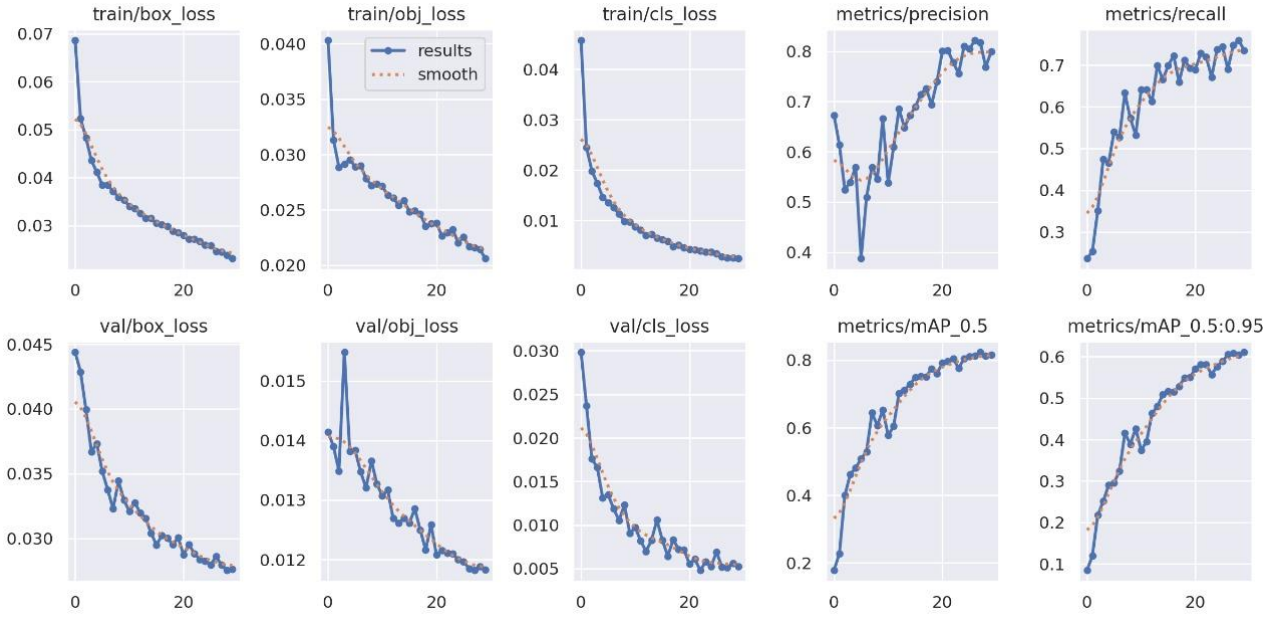
8. Features

- **Image Upload and Webcam Input:** Versatile data input for real-world applications.
- **Real-time Object Detection:** Fast and accurate classification of waste items.
- **Bounding Boxes with Labels:** Clearly marked waste items with confidence scores.
- **Downloadable Annotated Results:** Save the detection output for reference or compliance.
- **Recyclability Labeling:** Each item is marked as recyclable, non-recyclable, or compostable.
- **Secure File Handling:** Input files are processed within isolated directories to ensure data privacy.

DATASET CLASS DISTRIBUTION



GRAPHICAL RESULT



RESULTS



Waste Object Detection

Upload an image

Drag and drop file here
Limit 200MB per file • JPG, JPEG, PNG
Browse files

card.jpg 48.7KB

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the use_container_width parameter instead.



Uploaded Image

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the use_container_width parameter instead.



Prediction Result

Detected Waste Items:

- cardboard (Recyclable) 0.88

Download Annotated Image

9. Security Features

- **Input Handling:** Proper file type validation to avoid malicious uploads.
- **Scoped File Usage:** Files are only accessed within the project's working directory.
- **Streamlit Sandboxing:** Limits system exposure to only necessary resources.

10. Testing and Validation

- **Lighting Conditions:** Tested across different indoor and outdoor lighting scenarios to ensure robust detection.
- **Variety of Waste Items:** Validated with a wide range of materials – plastic, paper, metal, glass, and more.
- **Confidence Thresholds:** Tuned for optimal balance between false positives and missed detections.
- **Platforms:** Verified performance both on Google Colab (GPU-accelerated) and local CPU environments.

11. Conclusion

The **Smart Waste Classifier** project demonstrates the application of modern deep learning models (YOLOv5) for automating waste classification. By integrating object detection with a streamlined web interface, the system empowers individuals and organizations to practice effective waste segregation effortlessly. The project's **scalable and extendable** design also makes it suitable for integration with **smart bins** or public waste management infrastructure, contributing to **smart city initiatives** and **sustainability goals**.

12. Future Enhancements

- **IoT Smart Bin Integration:** Connect to physical smart bins for real-time waste classification at collection points.
- **User Waste Tracking:** Maintain user-specific data logs for behavior analysis and gamification.
- **Dataset Expansion:** Incorporate additional waste types for broader classification coverage.
- **Mobile App Interface:** Develop a mobile-friendly UI for convenience and accessibility.
- **Audio Feedback:** Enhance accessibility with spoken feedback for users with visual impairments.
- **Model Improvements:** Explore advanced models like YOLOv8 or Transformer-based detection for even higher accuracy.

13. Resources

https://github.com/Zzabi/Waste_Detection_YOLOv5/tree/main

https://youtu.be/DdeXrsMFnq4?si=npzBvj45ES_a84f5